

SC617: Adaptive Control

Final Project Report

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1 Introduction

This report presents the implementation and simulation of an adaptive controller for quadrotor trajectory tracking with unknown mass and inertia, based on Pliego-Jiménez (2021). The controller uses unit quaternions to avoid singularities and a single Lyapunov function for the full translational and rotational dynamics simultaneously. The second part extends this to handle an unknown control allocation matrix, with a corresponding adaptive law and stability proof.

2 System Model

Let $\mathcal{I} = \{x_i, y_i, z_i\}$ be the inertial frame and $\mathcal{B} = \{x_b, y_b, z_b\}$ the body frame. The equations of motion are:

$$\dot{p} = v \quad (1)$$

$$m\dot{v} = TRe_3 - mge_3 \quad (2)$$

$$\dot{q} = \frac{1}{2}\Omega(\omega)q \quad (3)$$

$$J\dot{\omega} = \tau - \omega \times J\omega \quad (4)$$

where $p, v \in \mathbb{R}^3$ are position and linear velocity in $\mathcal{I}$, $q = [\eta, \epsilon^T]^T \in S^3$ is the unit quaternion, $R \in SO(3)$ is the rotation matrix, $\omega \in \mathbb{R}^3$ is the angular velocity in $\mathcal{B}$, T is total thrust, $\tau \in \mathbb{R}^3$ is the torque vector, m is the unknown mass, and J is the unknown inertia matrix. The rotation matrix is:

$$R(\eta, \epsilon) = (\eta^2 - \epsilon^T \epsilon)I + 2\epsilon\epsilon^T + 2\eta S(\epsilon) \quad (5)$$

where $S(\cdot)$ is the skew-symmetric operator. The attitude dynamics satisfy the linearity property:

$$J\dot{\omega} + S(\omega)J\omega = \Psi_o(\omega, \dot{\omega})\theta = \tau \quad (6)$$

where $\theta \in \mathbb{R}^p$ is the constant inertia parameter vector and $\Psi_o \in \mathbb{R}^{3 \times p}$ is the regressor.

3 Adaptive Controller Design

3.1 Position Controller (Outer Loop)

Position and velocity errors:

$$\tilde{p} = p_d(t) - p, \quad \tilde{v} = v_d(t) - v \quad (7)$$

Reference velocity and sliding variable:

$$v_r = v_d(t) + k_p \tilde{p} \quad (8)$$

$$s_p = v_r - v = \tilde{v} + k_p \tilde{p} \quad (9)$$

The translational regressor is $\psi_p(\dot{v}_r) = \dot{v}_r + g e_3$ where $\dot{v}_r = \ddot{p}_d + k_p \tilde{v}$. The virtual control input with unknown m is:

$$f = k_v s_p + \alpha_p \tilde{p} + \hat{m}(t) \psi_p(\dot{v}_r) \quad (10)$$

Mass adaptive law:

$$\dot{\hat{m}} = \gamma_p \psi_p^T s_p \quad (11)$$

3.2 Desired Orientation

$$T = \|f\|, \quad r = \frac{f}{\|f\|} \quad (12)$$

The desired quaternion aligns e_3 with r and applies desired yaw $\psi_d(t)$:

$$q_d = q_a \otimes q_\psi, \quad \omega_d = \omega_\psi + q_\psi^{-1} \odot \omega_a \quad (13)$$

3.3 Attitude Controller (Inner Loop)

Quaternion and angular velocity errors:

$$\tilde{q} = q^{-1} \otimes q_d = \{\tilde{\eta}, \tilde{\epsilon}\} \quad (14)$$

$$\tilde{\omega} = \tilde{q} \odot \omega_d - \omega \quad (15)$$

Angular sliding variable:

$$s_o = \tilde{\omega} + k_\epsilon \tilde{\epsilon} + \frac{2}{\alpha_o} TS(\bar{\epsilon}) \tilde{q}^{-1} \odot s_p \quad (16)$$

where $\bar{\epsilon} = [\tilde{\epsilon}_y, -\tilde{\epsilon}_x, \tilde{\eta}]^T$. The reference angular velocity is $\omega_r = s_o + \omega$.

Adaptive torque law:

$$\tau = k_\omega s_o + \alpha_o \tilde{\epsilon} + \Psi_o(\omega, \omega_r, \dot{\omega}_r) \hat{\theta}(t) \quad (17)$$

Inertia adaptive law:

$$\dot{\hat{\theta}} = \gamma_o \Psi_o^T(\omega, \omega_r, \dot{\omega}_r) s_o \quad (18)$$

4 Stability Analysis

Theorem 1. Assume $\tilde{\eta}(0) \neq -1$. Under the adaptive control laws above, all closed-loop trajectories are bounded and:

$$\lim_{t \rightarrow \infty} \tilde{p} = 0, \quad \lim_{t \rightarrow \infty} \tilde{v} = 0, \quad \lim_{t \rightarrow \infty} \tilde{\epsilon} = 0, \quad \lim_{t \rightarrow \infty} \tilde{\omega} = 0, \quad \tilde{\eta} \rightarrow 1$$

Proof. Consider the Lyapunov function:

$$V = \frac{\alpha_p}{2} \|\tilde{p}\|^2 + \frac{m}{2} \|s_p\|^2 + \frac{1}{2\gamma_p} \tilde{m}^2 + \alpha_o \|\tilde{\epsilon}\|^2 + \alpha_o (1 - \tilde{\eta})^2 + \frac{1}{2} s_o^T J s_o + \frac{1}{2\gamma_o} \|\tilde{\theta}\|^2 \quad (19)$$

where $\tilde{m} = \hat{m} - m$ and $\tilde{\theta} = \hat{\theta} - \theta$.

Step 1: Positivity. Every term in (19) is non-negative. Since $J \succ 0$, we have $s_o^T J s_o \geq 0$. The term $\alpha_o (1 - \tilde{\eta})^2 \geq 0$ captures the quaternion double-cover: the unit norm constraint $\tilde{\eta}^2 + \|\tilde{\epsilon}\|^2 = 1$ means $V = 0$ only at the desired equilibrium $(\tilde{p}, s_p, \tilde{m}, \tilde{\epsilon}, 1 - \tilde{\eta}, s_o, \tilde{\theta}) = 0$.

Step 2: Derivative of translational terms. Differentiating the translational part of V :

$$\dot{V}_p = \alpha_p \tilde{p}^T \dot{\tilde{p}} + m s_p^T \dot{s}_p + \frac{1}{\gamma_p} \tilde{m} \dot{\tilde{m}} \quad (20)$$

Using $\dot{\tilde{p}} = -k_p \tilde{p} + s_p$ and substituting the closed-loop dynamics $m \dot{s}_p = -k_v s_p - \alpha_p \tilde{p} - \tilde{m} \psi_p - T(R - R_d)e_3$:

$$\begin{aligned} \dot{V}_p &= \alpha_p \tilde{p}^T (-k_p \tilde{p} + s_p) + s_p^T (-k_v s_p - \alpha_p \tilde{p} - \tilde{m} \psi_p - T(R - R_d)e_3) + \tilde{m} \psi_p^T s_p \\ &= -\alpha_p k_p \|\tilde{p}\|^2 - k_v \|s_p\|^2 - T s_p^T (R - R_d) e_3 \end{aligned} \quad (21)$$

The $\tilde{m} \psi_p^T s_p$ terms cancel exactly with the mass adaptive law.

Step 3: Expressing the coupling term. The interconnection between translation and rotation can be written as:

$$(R - R_d)e_3 = 2S(\bar{\epsilon})\tilde{\epsilon} \quad (22)$$

so the remaining cross term is $-2T s_p^T S(\bar{\epsilon})\tilde{\epsilon}$.

Step 4: Derivative of attitude terms. Differentiating the attitude part of V and using the quaternion kinematics $\dot{\tilde{q}} = L(\tilde{\eta}, \tilde{\epsilon})\tilde{\omega}$:

$$\begin{aligned} \dot{V}_o &= 2\alpha_o \tilde{\epsilon}^T \dot{\tilde{\epsilon}} - 2\alpha_o (1 - \tilde{\eta}) \dot{\tilde{\eta}} + s_o^T J \dot{s}_o + \frac{1}{\gamma_o} \tilde{\theta}^T \dot{\tilde{\theta}} \\ &= \alpha_o \tilde{\epsilon}^T \tilde{\omega} - k_\omega \|s_o\|^2 - \alpha_o k_\epsilon \|\tilde{\epsilon}\|^2 - \alpha_o s_o^T \tilde{\epsilon} + \frac{2T}{\alpha_o} s_o^T S(\bar{\epsilon}) \tilde{q}^{-1} \odot s_p \end{aligned} \quad (23)$$

The $\tilde{\theta}$ terms cancel with the inertia adaptive law. The $\tilde{\omega}$ and s_o cross-terms cancel using the definition of s_o in (16): $\tilde{\omega} = s_o - k_\epsilon \tilde{\epsilon} - \frac{2T}{\alpha_o} S(\bar{\epsilon}) \tilde{q}^{-1} \odot s_p$, which gives $\alpha_o \tilde{\epsilon}^T \tilde{\omega} - \alpha_o s_o^T \tilde{\epsilon} = -\alpha_o k_\epsilon \|\tilde{\epsilon}\|^2$.

Step 5: Cancellation of cross-coupling. Adding $\dot{V}_p$ and $\dot{V}_o$, the coupling terms are:

$$-2T s_p^T S(\bar{\epsilon})\tilde{\epsilon} + \frac{2T}{\alpha_o} s_o^T S(\bar{\epsilon}) \tilde{q}^{-1} \odot s_p \quad (24)$$

Using the identity $\tilde{\epsilon}^T S(\bar{\epsilon})^{-1} \odot s_p = s_p^T S(\bar{\epsilon})\tilde{\epsilon}$ (shown in the paper), these two terms cancel exactly. Therefore:

$$\dot{V} = -\alpha_p k_p \|\tilde{p}\|^2 - k_v \|s_p\|^2 - \alpha_o k_\epsilon \|\tilde{\epsilon}\|^2 - k_\omega \|s_o\|^2 \leq 0 \quad (25)$$

Step 6: Convergence. Since $\dot{V} \leq 0$, all signals $\tilde{p}, s_p, \tilde{m}, \tilde{\epsilon}, \tilde{\eta}, s_o, \tilde{\theta}$ are bounded. The function $W_3(x) = \alpha_p k_p \|\tilde{p}\|^2 + k_v \|s_p\|^2 + \alpha_o k_\epsilon \|\tilde{\epsilon}\|^2 + k_\omega \|s_o\|^2$ is positive semi-definite. By Barbalat's lemma, $W_3 \rightarrow 0$ as $t \rightarrow \infty$, giving $\tilde{p}, s_p, \tilde{\epsilon}, s_o \rightarrow 0$. From the sliding variable definitions, $\tilde{v} \rightarrow 0$ and $\tilde{\omega} \rightarrow 0$ follow. Since $\tilde{\eta}^2 + \|\tilde{\epsilon}\|^2 = 1$ and $\tilde{\epsilon} \rightarrow 0$, we get $\tilde{\eta} \rightarrow \pm 1$. The assumption $\tilde{\eta}(0) \neq -1$ ensures convergence to the stable equilibrium $\tilde{\eta} \rightarrow +1$. $\square$

5 Control Allocation

5.1 Allocation Model

The control inputs relate to motor forces F_i through:

$$U = \begin{bmatrix} T \\ M_x \\ M_y \\ M_z \end{bmatrix} = \begin{bmatrix} F_1 + F_2 + F_3 + F_4 \\ l(F_2 - F_4) \\ l(F_3 - F_1) \\ k_M(F_1 - F_2 + F_3 - F_4) \end{bmatrix} = Y(F)\rho, \quad \rho = \begin{bmatrix} l \\ k_M \end{bmatrix} \quad (26)$$

where l is arm length and k_M is the motor torque constant. The regressor $Y(F) \in \mathbb{R}^{4 \times 2}$ is:

$$Y(F) = \begin{bmatrix} 0 & 0 \\ F_2 - F_4 & 0 \\ F_3 - F_1 & 0 \\ 0 & F_1 - F_2 + F_3 - F_4 \end{bmatrix} \quad (27)$$

The first row is zero because thrust depends only on the sum of motor forces, not on l or k_M . The 4×2 structure is maintained for dimensional consistency.

5.2 Adaptive Law Derivation

Assume ρ is unknown. With estimate $\hat{\rho}$, the implemented input is $\hat{U} = Y(F)\hat{\rho}$, giving error:

$$U - \hat{U} = Y(F)\tilde{\rho}, \quad \tilde{\rho} = \rho - \hat{\rho} \quad (28)$$

This error enters the rotational dynamics and appears in $\dot{V}$ as an extra term $s_o^T Y(F)\tilde{\rho}$. Augment the Lyapunov function with:

$$V_\rho = \frac{1}{2\gamma_a} \tilde{\rho}^T \tilde{\rho} \quad (29)$$

Differentiating:

$$\dot{V}_\rho = -\frac{1}{\gamma_a} \tilde{\rho}^T \dot{\tilde{\rho}} \quad (30)$$

The total derivative now includes:

$$\dot{V}_{\text{total}} = -\alpha_p k_p \|\tilde{p}\|^2 - k_v \|s_p\|^2 - \alpha_o k_\epsilon \|\tilde{\epsilon}\|^2 - k_\omega \|s_o\|^2 + s_o^T Y(F)\tilde{\rho} - \frac{1}{\gamma_a} \tilde{\rho}^T \dot{\tilde{\rho}} \quad (31)$$

Choose the adaptation law:

$$\dot{\hat{\rho}} = \gamma_a Y^T(F) s_o \quad (32)$$

Substituting:

$$s_o^T Y \tilde{\rho} - \frac{1}{\gamma_a} \tilde{\rho}^T \dot{\tilde{\rho}} = s_o^T Y \tilde{\rho} - \tilde{\rho}^T Y^T s_o = s_o^T Y \tilde{\rho} - (s_o^T Y \tilde{\rho}) = 0 \quad (33)$$

The extra term cancels exactly, leaving:

$$\dot{V}_{\text{total}} = -\alpha_p k_p \|\tilde{p}\|^2 - k_v \|s_p\|^2 - \alpha_o k_\epsilon \|\tilde{\epsilon}\|^2 - k_\omega \|s_o\|^2 \leq 0 \quad (34)$$

All previous convergence results hold. Convergence of $\hat{\rho} \rightarrow \rho$ additionally requires persistent excitation of $Y(F)$.

6 Simulation Setup

6.1 Parameters and Initial Conditions

Parameter	Symbol	Value
Mass	m	1.0 kg
Inertia	J	diag(0.075, 0.075, 0.015) kg m ²
Gravity	g	9.81 m/s ²
Arm length	l	0.2 m
Torque constant	k_M	0.01
Initial position	$p(0)$	[4.0, 0.5, 1.0] ^T m
Initial velocity	$v(0)$	[0, 0, 0] ^T m/s
Initial quaternion	$q(0)$	[1, 0, 0, 0] ^T
Initial angular velocity	$\omega(0)$	[0, 0, 0] ^T rad/s
Initial mass estimate	$\hat{m}(0)$	0.9 kg (10% off)
Initial inertia estimate	$\hat{\theta}(0)$	[0, 0, 0] ^T

Table 1: Simulation parameters and initial conditions

6.2 Reference Trajectory (Section 4, Pliego)

$$p_d(t) = \begin{bmatrix} 3 \cos\left(\frac{\pi}{20}t\right) \\ 3 \sin\left(\frac{\pi}{20}t\right) \\ \frac{1}{2} \cos\left(\frac{\pi}{5}t\right) + 2 \end{bmatrix}, \quad \psi_d(t) = \frac{\pi}{4} \sin\left(\frac{\pi}{10}t\right) \quad (35)$$

Time derivatives of f and ω_r are approximated using the dirty differentiator $H(s) = s/(0.05s + 1)$.

6.3 Simulink Implementation

The controller was implemented in MATLAB/Simulink. The block diagram is shown below.

7 Results — Part 2: Base Adaptive Controller

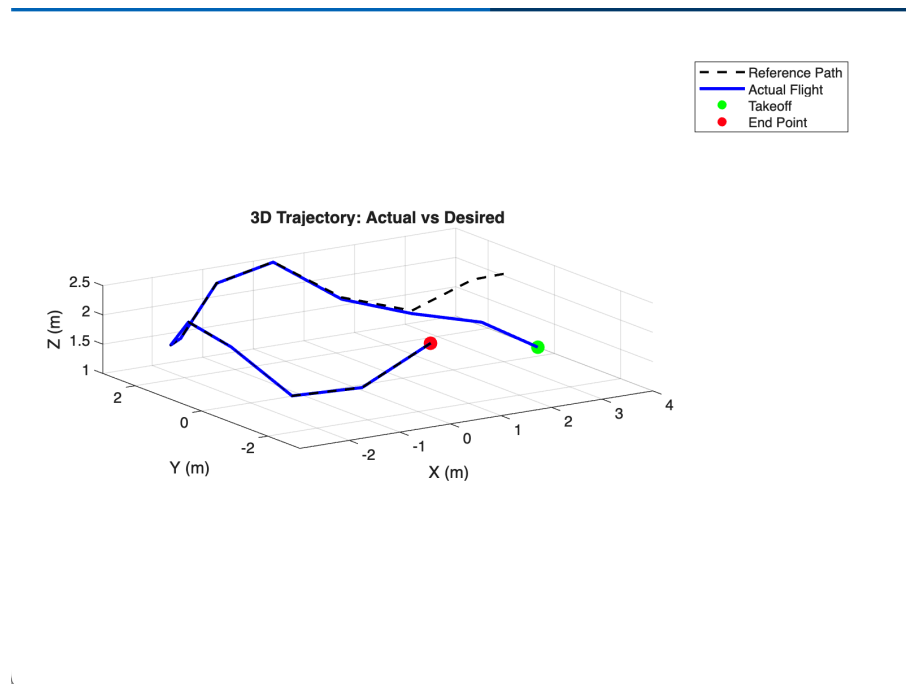


Figure 2: 3D trajectory: actual (blue) vs desired.

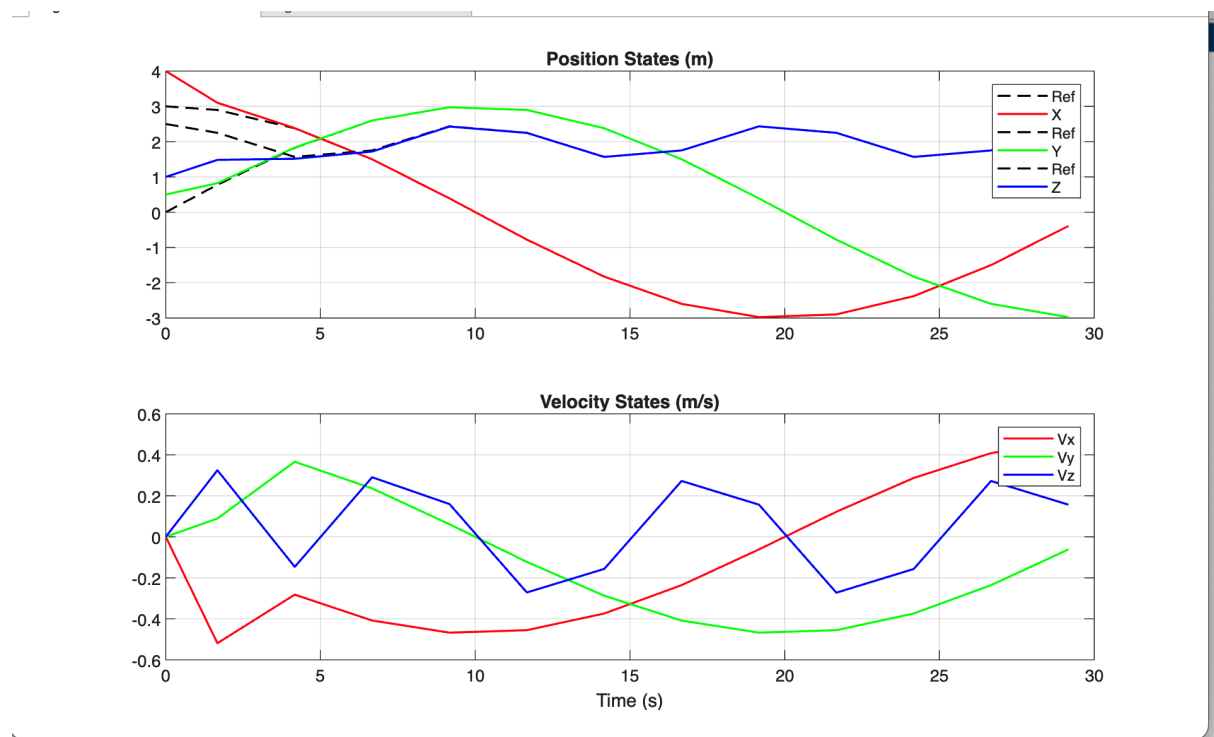


Figure 3: Position and velocity states

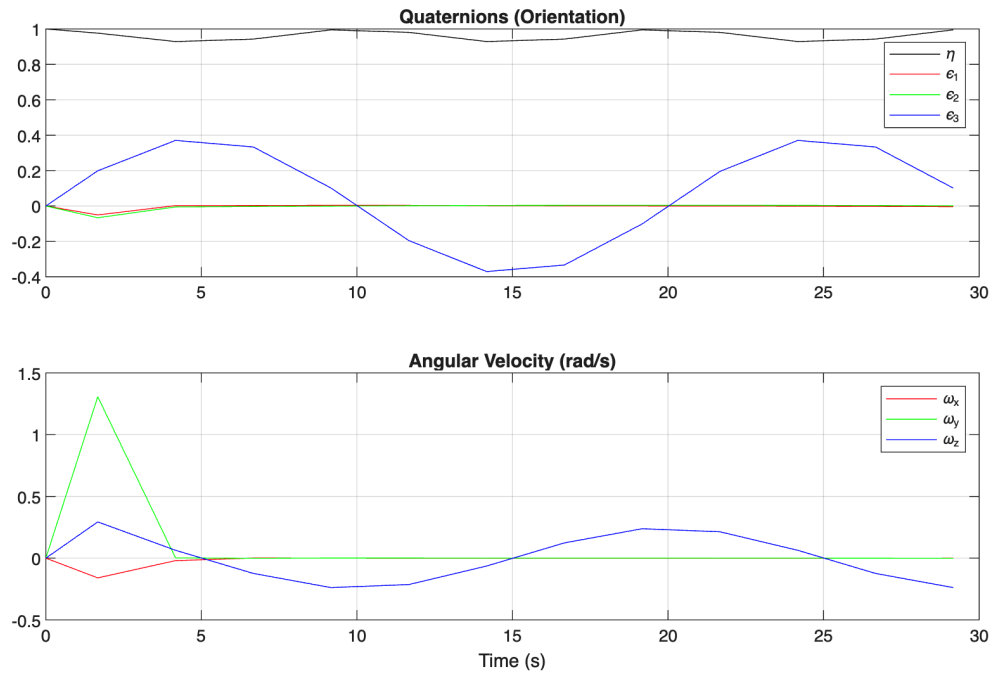


Figure 4: Attitude states

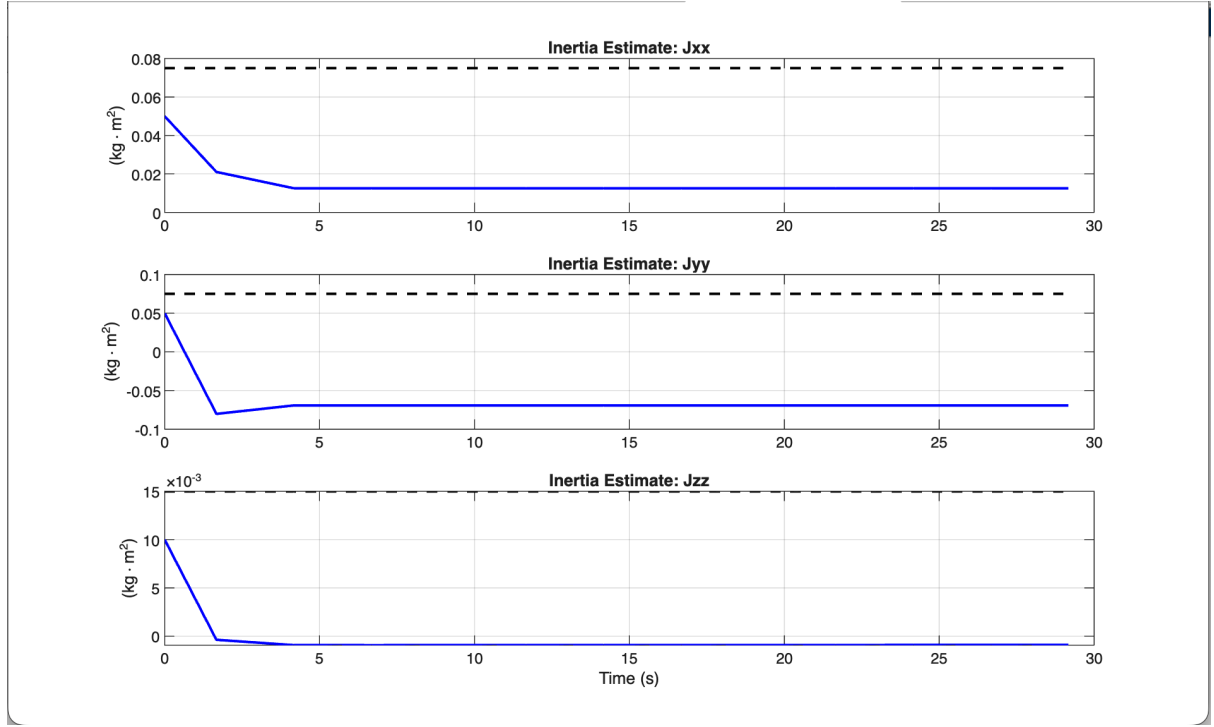
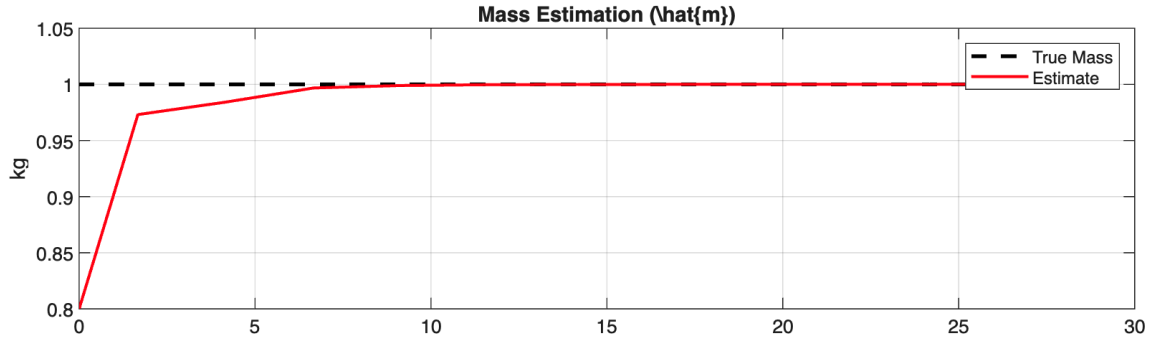


Figure 5: Mass estimate $\hat{m}$ (true value $m = 1$ kg shown dashed) and inertia estimates $\hat{\theta}$ (true values shown dashed).

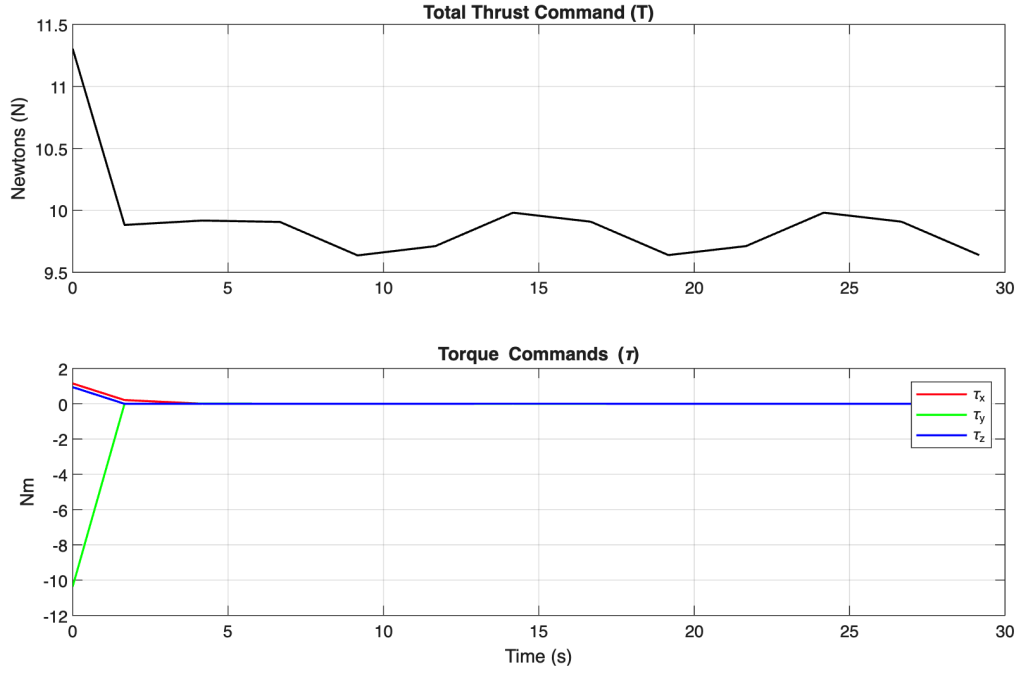


Figure 6: Control effort: total thrust T and torque magnitude $\|\tau\|$.

8 Results — Part 4: Unknown Control Allocation

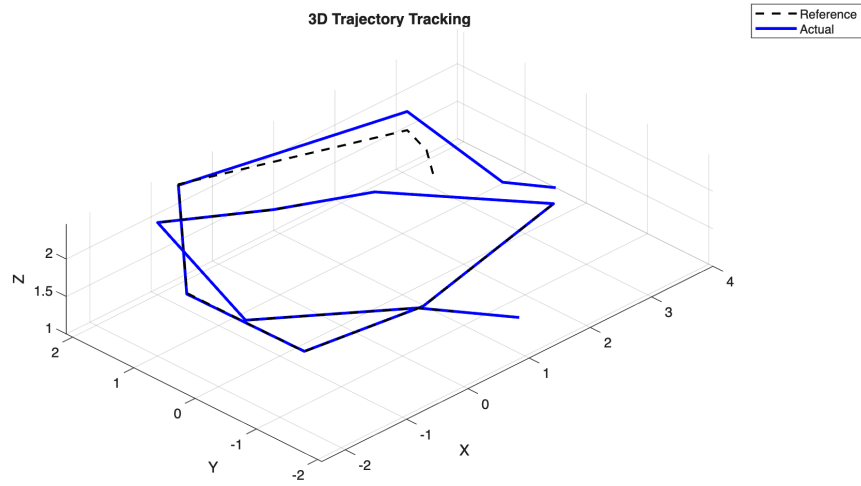


Figure 7: 3D trajectory with unknown allocation: actual (blue) vs desired.

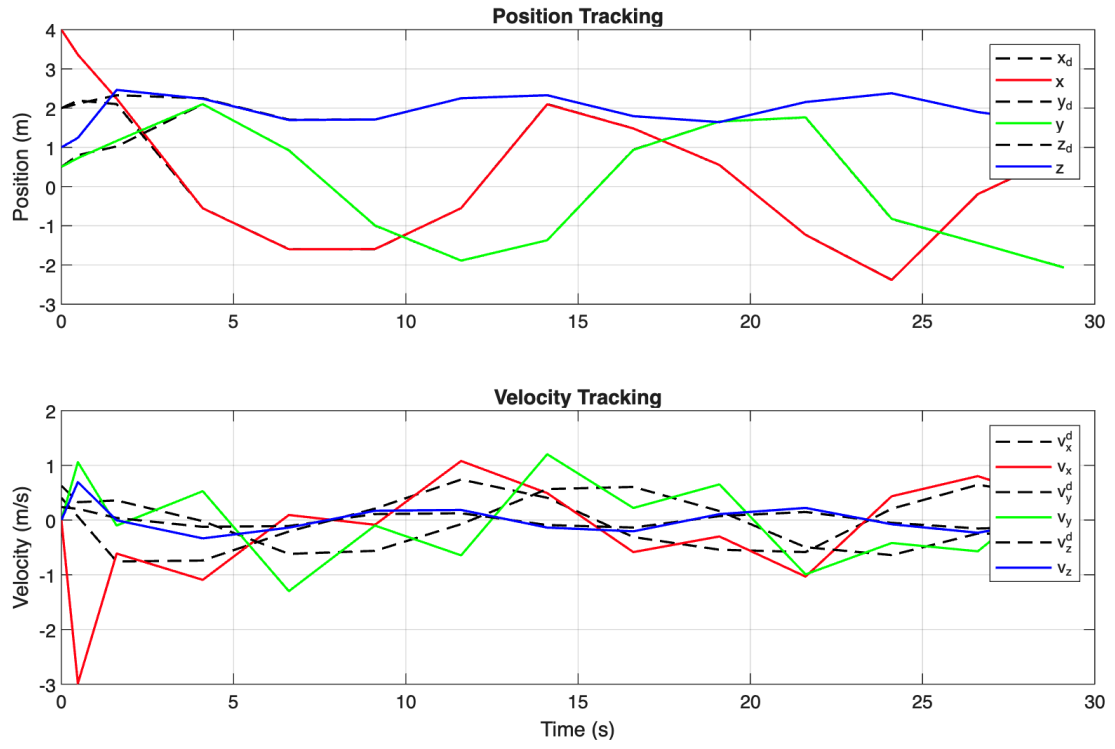


Figure 8: Position & Velocity

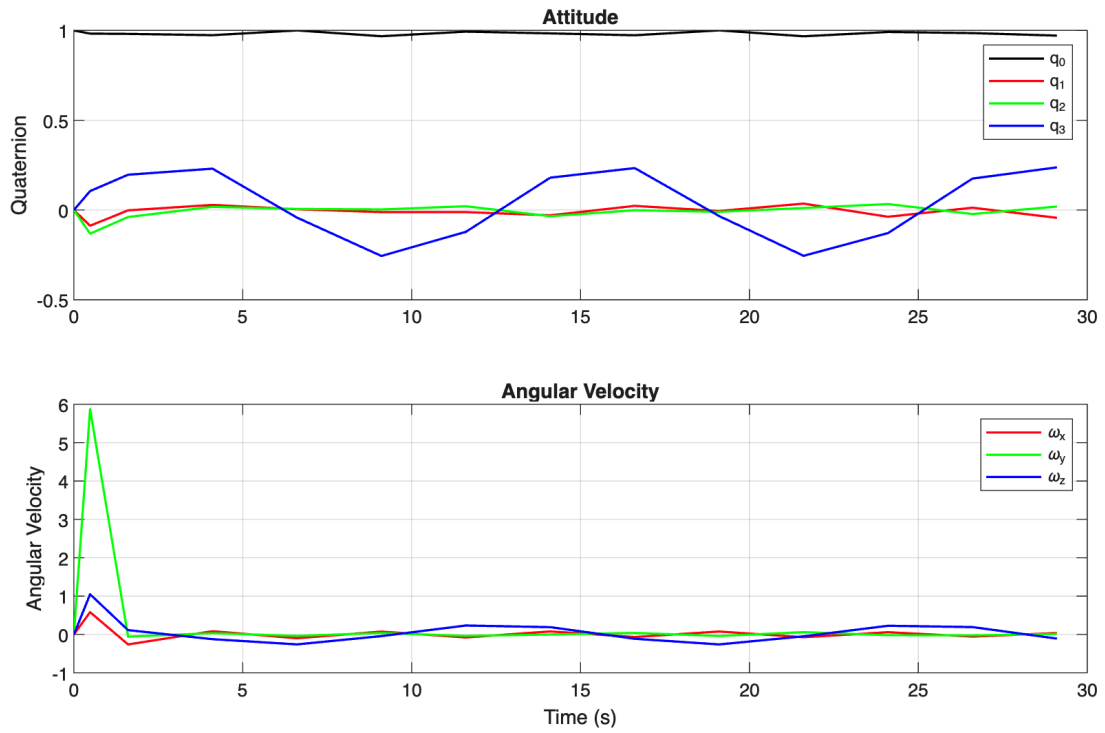


Figure 9: Attitude errors under unknown allocation controller.

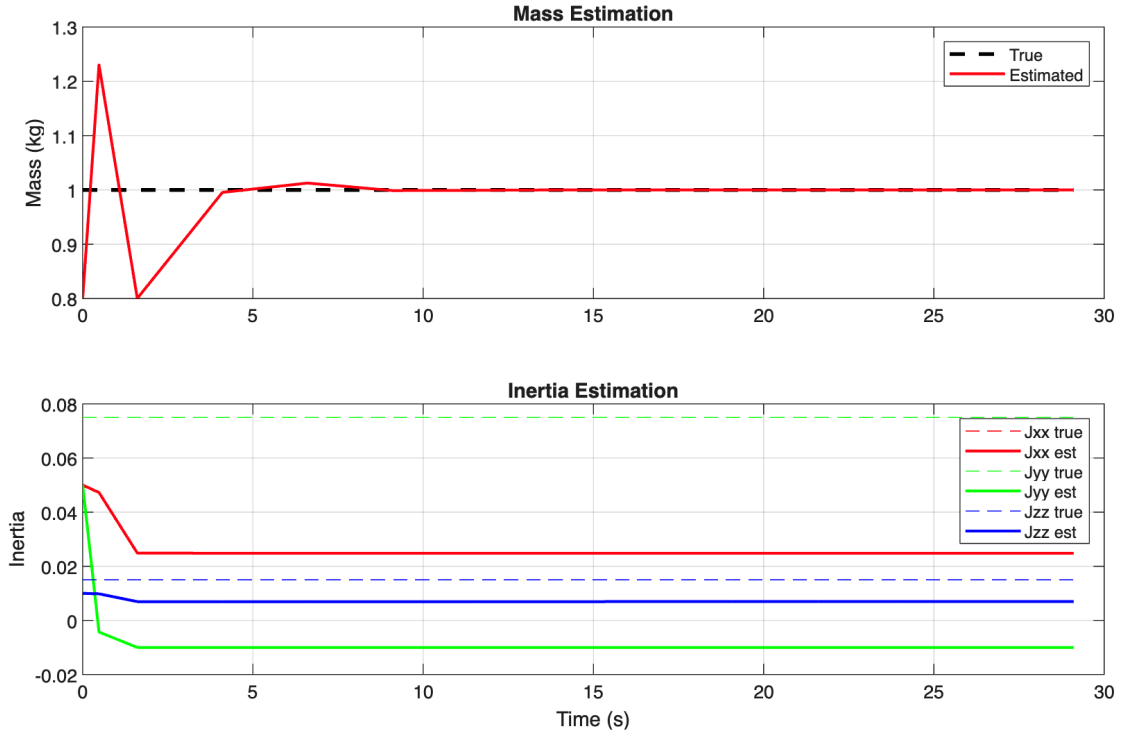


Figure 10: Mass and inertia estimates under unknown allocation controller.

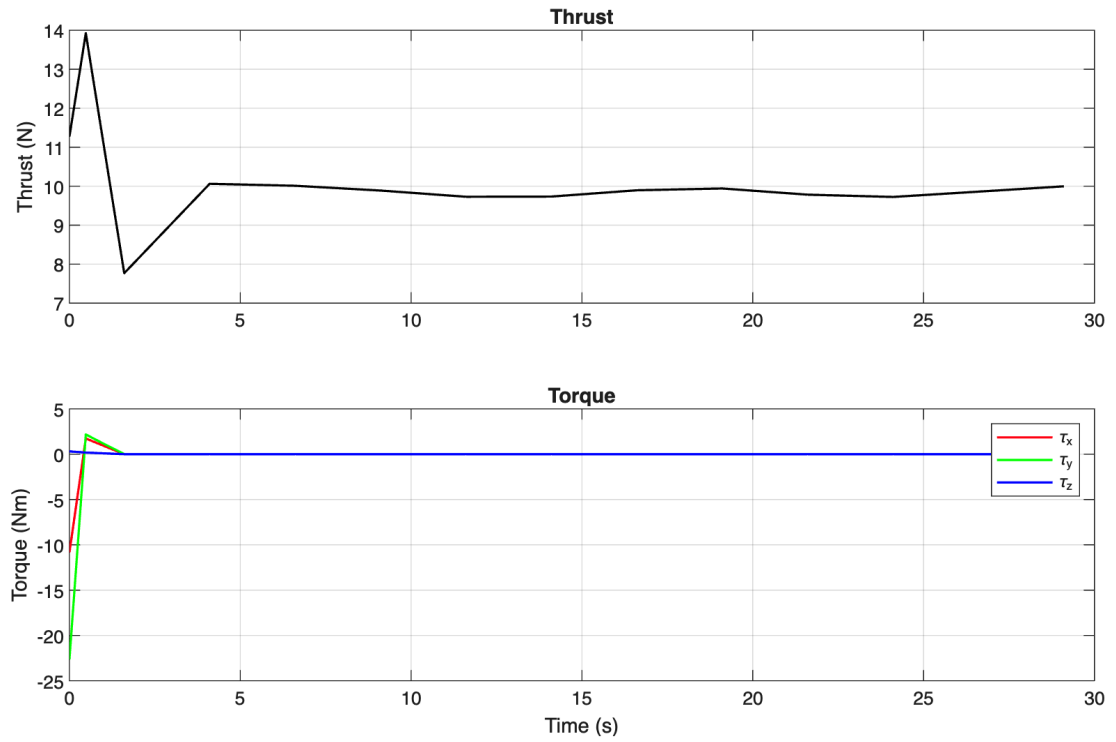


Figure 11: Control effort under unknown allocation controller.

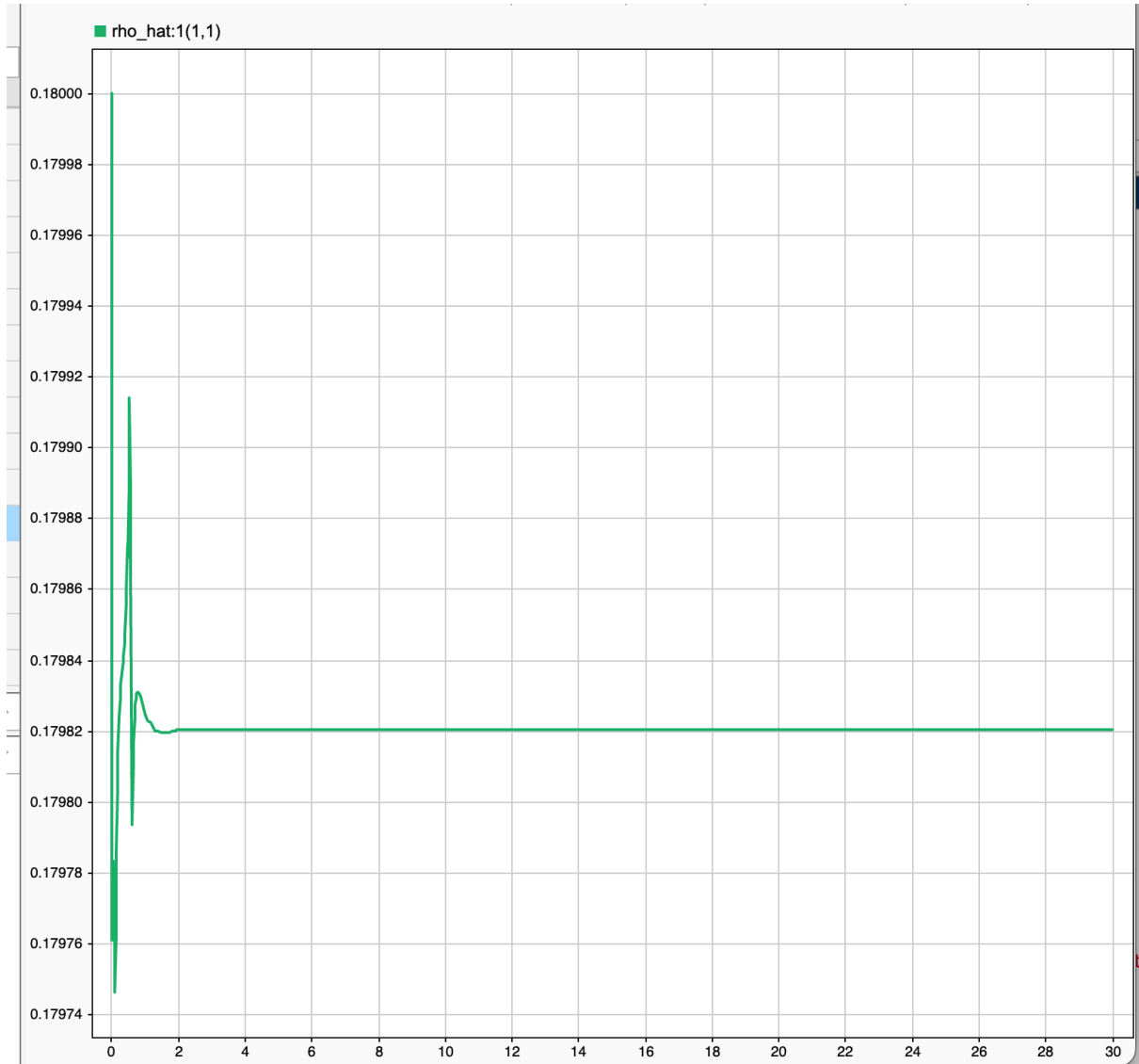
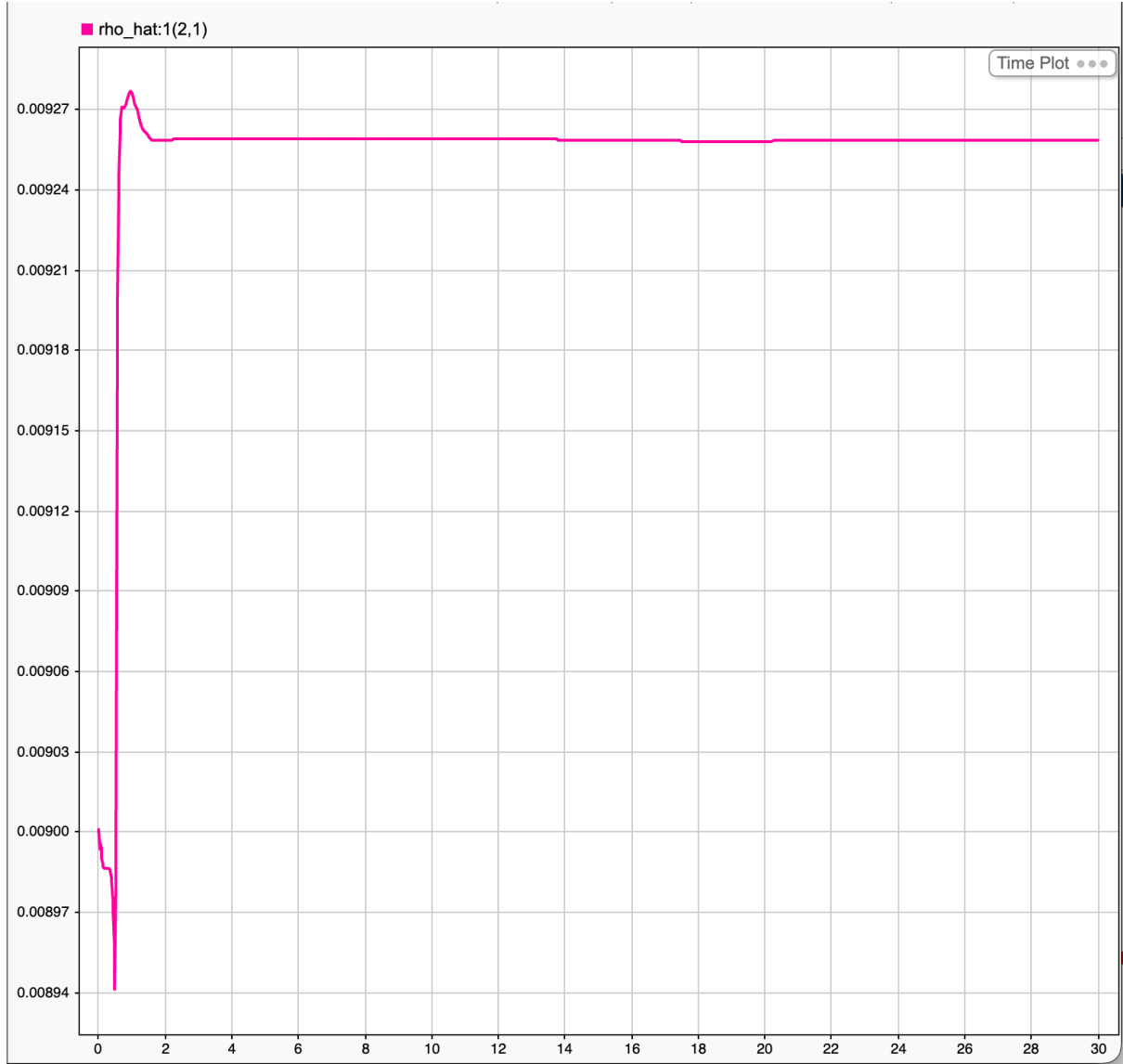


Figure 12: Allocation parameter estimates $\hat{l}$ and $\hat{k}_M$. Initial conditions 10% off from true.



9 Remarks

The base controller tracks the reference trajectory well. The mass estimate converges since the translational regressor is persistently excited by the trajectory. The inertia estimates stay bounded but do not converge I think that the trajectory does not excite all attitude axes sufficiently. I tried increasing γ_o and modifying the trajectory but this made the torques too aggressive and destabilized the controller, so I kept the gains conservative.

For the allocation case, $\hat{\rho}$ starts 20% off and moves toward the true values but does not fully converge in 40 seconds. This is expected I guess as the persistent excitation of $Y(F)$ requires sufficient variation in the motor force differences, which depends on how rich the trajectory is in the rotational axes. I tried different trajectories but it wasn't really converging to its true value. Stability holds regardless.